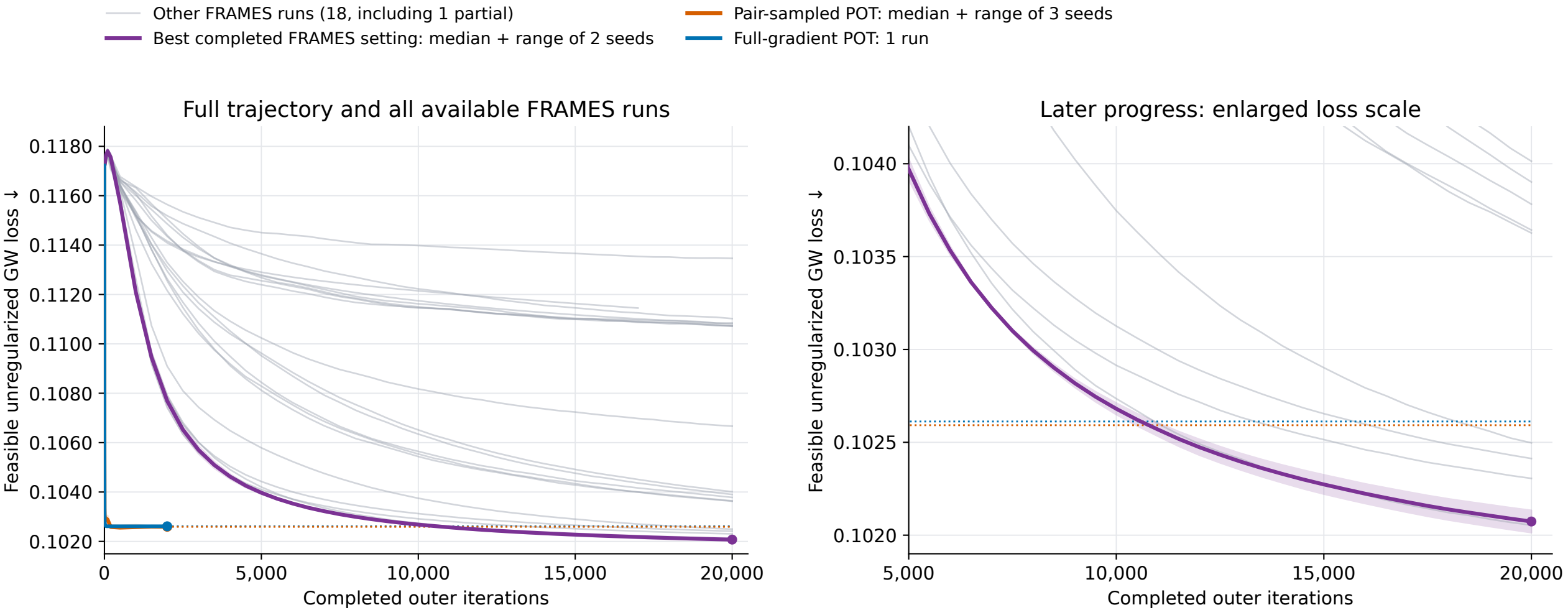
SHARE-seq: loss versus iterations

31,890 cells per modality · FP64 · Product initialization · Observations saved by 3 November 2026



FRAMES: 1,024 sampled pairs/update; best setting γ_0=0.3, ρ_0=0.1, smoothing scale=10. Theoretical exponents fixed.

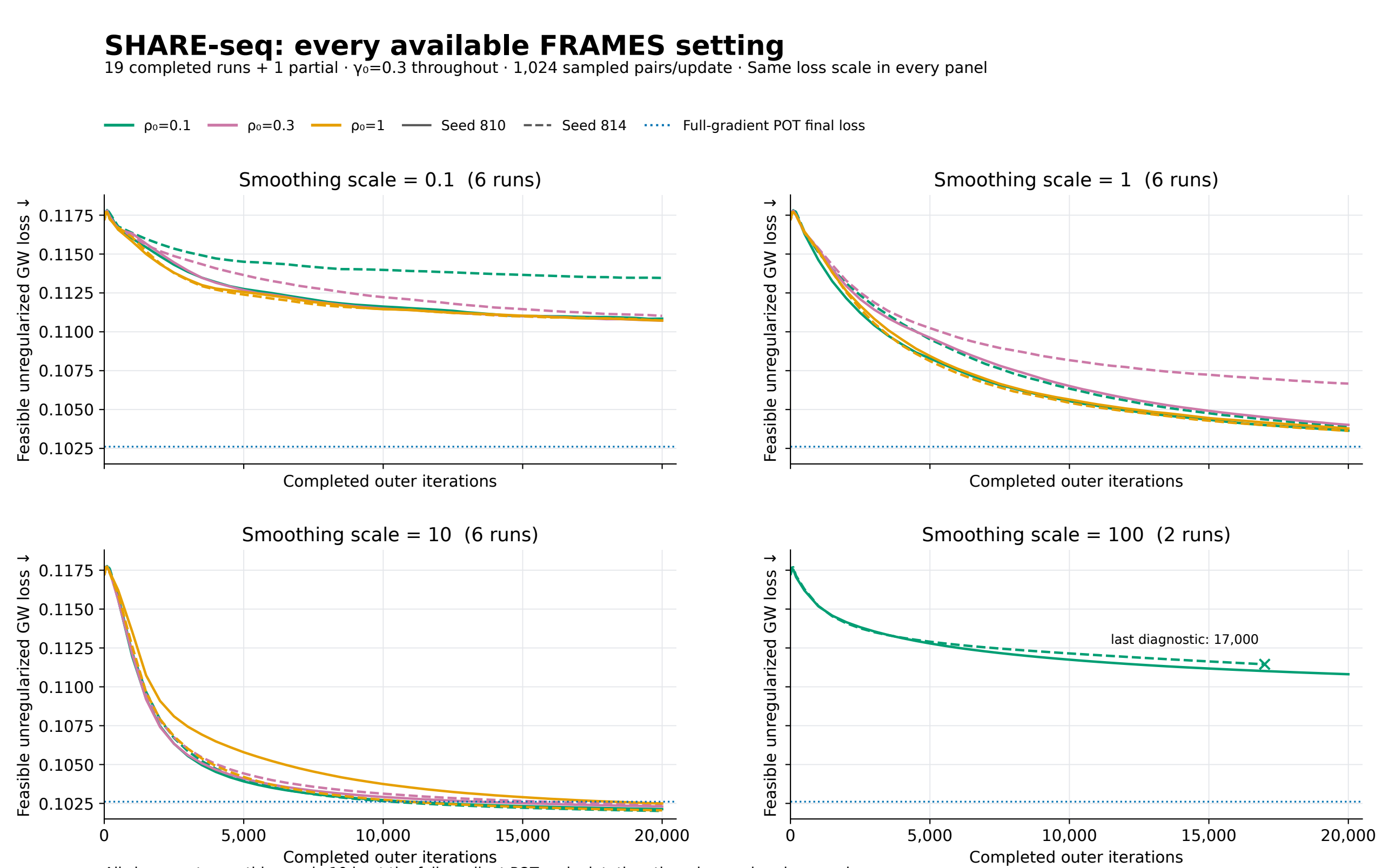
POT: $\varepsilon=10^{-3}$, step=$1/\varepsilon$; sampled version uses 1,024 pairs/update. POT trajectories end at 2,000 updates.

Dotted segments beyond 2,000 are final POT loss references, not additional measured iterations. Bands show seed ranges.

FRAMES setting selected on these tuning seeds; fresh-seed confirmation is pending. Lines connect saved diagnostics.

SHARE-seq: every available FRAMES setting

19 completed runs + 1 partial · γ_0=0.3 throughout · 1,024 sampled pairs/update · Same loss scale in every panel



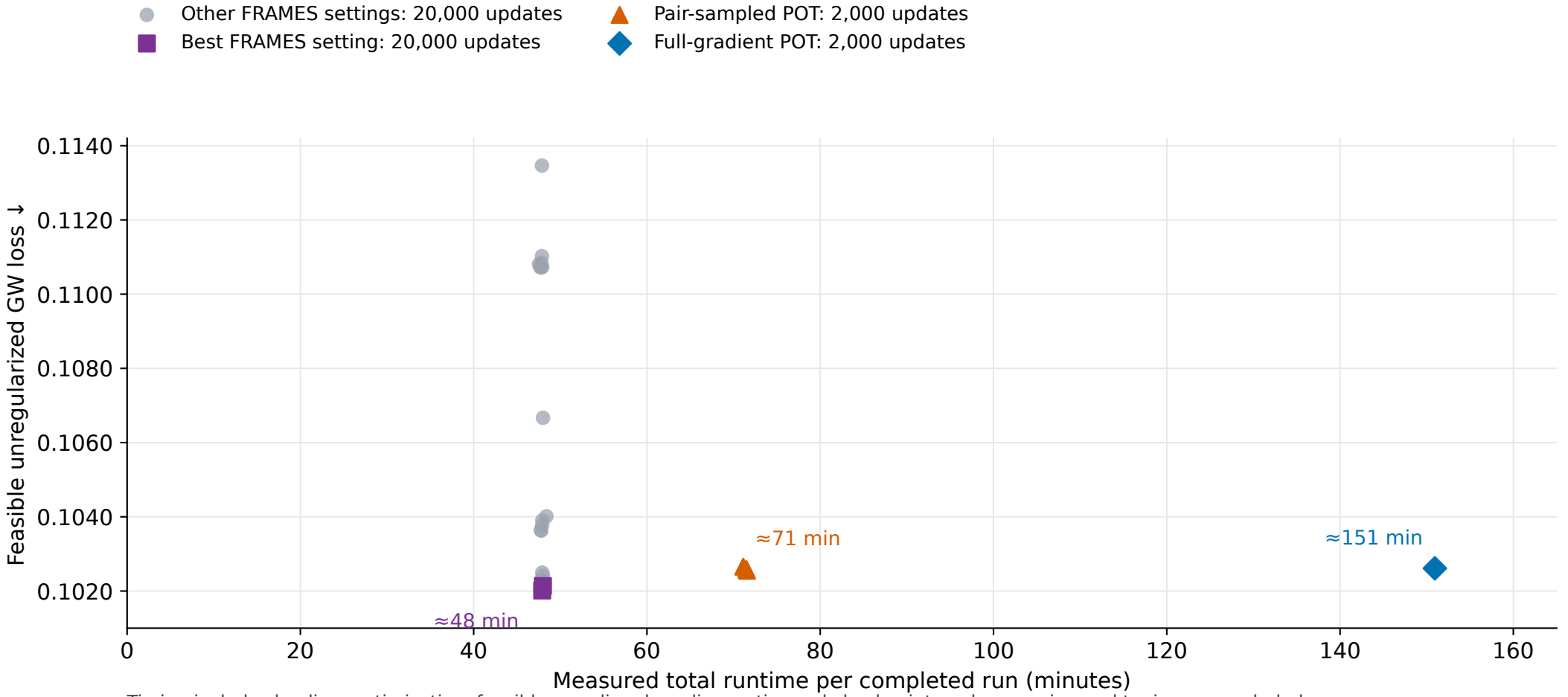
All six runs at smoothing scale 10 beat the full-gradient POT endpoint; the other observed scales remain worse.

Scale 100 has only ρ_0=0.1 available; one run is saved at 17,451 updates, with its last loss diagnostic at 17,000.

The γ_0=1 grid and remaining γ_0=0.3 settings have not produced results. No smoothing or loss values are extrapolated.

SHARE-seq: final loss versus measured total runtime

Completed runs only · Four A100 GPUs per run · Endpoints, not loss-versus-time trajectories



Timing includes loading, optimization, feasible rounding, loss diagnostics and checkpoint work; queueing and tuning are excluded.

Per-diagnostic timestamps were not recorded. No interpolation of iteration counts into time is used.

These endpoints do not establish time to a target loss: full-gradient POT was already near its final loss by 50–200 updates.

All runs used fixed iteration budgets. FRAMES selection used the same tuning seeds shown here; confirmation is pending.